

BIL 481 Software Engineering

Project Assignment 1

Project Concept Development with Gemini

Released Friday, 2 October 2026

Due Friday, 9 October 2026

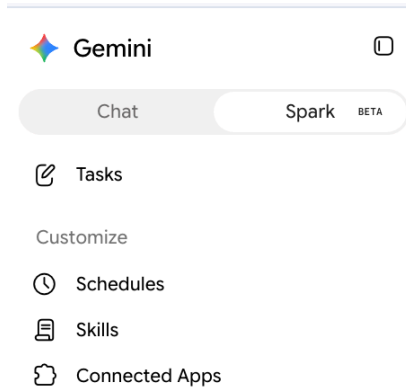
Instructor Dr. Çiğdem Avcı

TA Ömer Kılınç

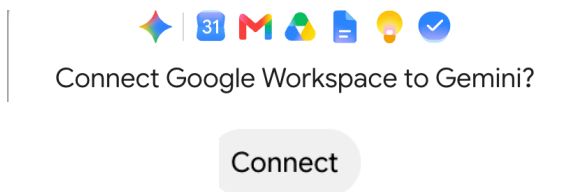
Overview. This is the first assignment of your course project. Working as a group, you will create **Concept Notes** outlining your project's problem, intended users, requirements, and preliminary design and implementation plan. **Gemini is central to this assignment:** use it to draft, critique, compare, and refine your proposal. Evaluate its suggestions together and make the final project decisions as a group.

1 Setup

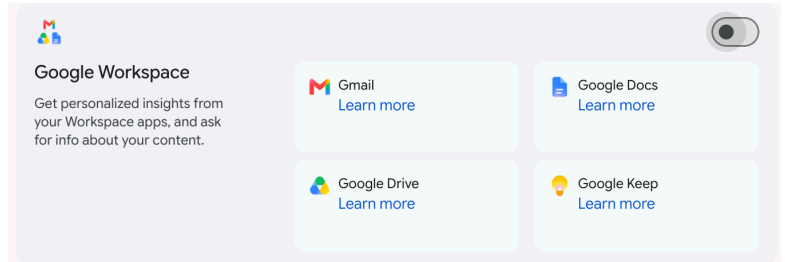
1. Sign in to [Gemini](#) and Google Drive using the **same Google account**, then select **Spark**.
2. Open **Connected Apps** from the left sidebar.
3. Turn on **Google Workspace**. Read the permission dialog and select **Connect** to allow the Drive access needed for this assignment.
4. Confirm that Google Workspace is enabled. Use Drive to store your Concept Notes document.



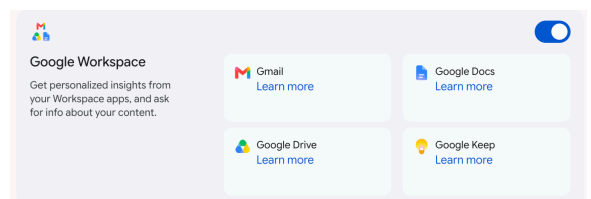
1. Select Spark and open Connected Apps.



3. Review the permissions and select Connect.



2. Turn on Google Workspace.



4. Check that the switch is enabled.

2 Write the initial prompt

In Google Drive, create a folder named BIL481-2026-Project for your group's Concept Notes and the files you will add as the project progresses.

Write one prompt describing your group's project idea and asking Gemini to brainstorm its overall structure, core concepts, and implementation approach. Include the problem, intended users, main goals, and any known constraints.

End the prompt with an explicit command asking Gemini to create a document titled **Concept Notes** in your **Google Drive project folder**, containing all seven sections below. Open the document in Drive to confirm that it was created.

3 Required Concept Notes sections

1. **Project Name.** Give the project a meaningful name.
2. **Short Description.** Briefly describe the problem, intended users, purpose, and main idea of the proposed solution.
3. **Requirements.** Include approximately **10 numbered requirements**, covering system behavior and relevant quality expectations. Make each requirement clear enough to discuss and check.
4. **Simple Pseudocode for Implementation Parts.** Show the basic logic of the main parts or workflows, including inputs, main steps, and expected outputs.
5. **Sample Views.** Include a few screens or views showing how users interact with the system. Sketches, wireframes, or clearly structured screen layouts are sufficient.
6. **Sample Design Information.** Give preliminary design details, such as main components, data entities, interactions, or technology choices. Explain how they support the concept.
7. **Scenarios.** Include a few concrete usage examples. Identify the user, goal, main steps, and expected outcome in each scenario, keeping them consistent with the requirements and views.

4 Critique, compare, and refine

Continue the same Gemini conversation and make decisions as a group.

Critique the proposal. Ask Gemini to identify weaknesses, unclear or missing requirements, design inconsistencies, and feasibility concerns. Discuss the critique and ask follow-up questions about the issues that matter to your project.

Compare similar applications. Ask Gemini to compare your concept with similar applications or approaches. Examine shared features, differences, and opportunities to improve the idea. Check any examples or claims you rely on, and keep the references in the conversation or document.

Refine the concept. Record which suggestions your group accepts, changes, or rejects, and explain why. **Use command prompts in Gemini to update the existing Concept Notes document in Drive.** For each accepted change, specify the document, project folder, section, and exact revision to apply. Retain all seven required sections, then open the document in Drive to verify that each update was saved.

Example command prompt: "Update the Requirements section of Concept Notes in our Drive project folder with the agreed changes, and save the document."

5 Initialize the project repository

Fork the template repository below, clone your fork locally, and ensure that all group members can access it.

<https://github.com/CPPPriest/BIL481-2026-Project>

As part of PA1, complete your fork's README.md with your group name and each member's full name, GitHub username, email, and school number. Replace the example entries, then commit and push the updated README.

Full Name	GitHub Username	Email	School Number
Ömer Kılınç (<i>example</i>)	@github-username	omerkilinc@etu.edu.tr	261111011
Member 2	@github-username	email@etu.edu.tr	School number
Member 3	@github-username	email@etu.edu.tr	School number
Member 4	@github-username	email@etu.edu.tr	School number
Member 5	@github-username	email@etu.edu.tr	School number

Template README: replace the example entries with your group's details.

6 Save and submit

Keep the Concept Notes document only in your project folder on Google Drive. Do not upload it elsewhere. Submit its Drive link and ensure that the course staff have viewing access. Open the document in Drive and verify that it is complete and readable.

Submission channel: To be announced. Submit the following items by the deadline on the first page, identifying the project and all group members clearly.

1. **Concept Notes (Google Drive link):** the refined document with all seven required sections and approximately 10 requirements, stored in your BIL481-2026-Project Drive folder in .docx or .md format.
2. **Complete Gemini chat history for PA1:** the exact initial prompt, initial response and Concept Notes, and every follow-up prompt and response used for critique, comparison, and refinement. Provide a readable transcript in conversation order, including generated document content when it is not visible in the transcript.
3. **Project repository link:** your group's fork with the completed README.md.

Before submitting, check that the requirements, pseudocode, views, design information, and scenarios describe the same project. The transcript should show how the concept evolved, and the final document should reflect your group's decisions. Every member should be able to explain the concept and its main design choices.

7 Questions and office hours

Use the course's **Piazza Q&A section** for assignment questions. You may also contact the TA directly by email or in person during the office hours below.

Teaching assistant Ömer Kılınç

Email omerkilinc@etu.edu.tr

Office hours Thursday, 16:30–18:20; Friday, 16:30–18:20

Location TM-213